

THENNANNAMALAI MALLIGARJUNAN

SOFTWARE ENGINEER | BACKEND & CLOUD-NATIVE SYSTEMS DEVELOPER

Dallas, TX | +1 945 527 5076 | thenn.dev@gmail.com | linkedin.com/in/thenn-annamalai-m/

EDUCATION

Master of Science in Computer Science | The University of Texas at Dallas, Richardson, TX | **GPA:** 3.74

May 2025

Bachelor's in Electrical and Electronic Engineering | Thiagarajar College of Engineering, India | **GPA:** 3.77

May 2021

PROFESSIONAL EXPERIENCE

Software Engineer | Zoho Corporation Private Ltd | Chennai, India

Mar 2021 – Aug 2023

Media Router

Media Router, a core Audio/Video infrastructure component, manages 100 million global clients, enabling real-time conferencing via WebRTC

- Designed, developed, and deployed high-reliability C++ applications, reducing media dispatch latency to under 150 ms for 100M+ users
- Introduced 60+ monitoring metrics, streamlined GDPR-compliant debug workflows, and resolved incidents 30% faster, improving resilience
- Led a team of three developers to implement temporal video scaling, decreasing call drops by 25% for low-bandwidth users
- Implemented load-balancing algorithms to distribute traffic across servers, reducing hot spots & lowering cloud costs by \$30K/year

TURN Server

TURN Server enables real-time conferencing in firewall-restricted networks while exposing clients' public IP addresses for global connectivity

- Enhanced a low-latency, concurrency-safe C++ message delivery system, improving distributed server communication throughput by 35%
- Integrated C++ applications with Java via JNI, enabling seamless cross-language operations and reducing integration errors by 25%
- Optimized network traffic using analysis tools and cryptographic libraries, maintaining <100 ms latency while ensuring secure delivery
- Applied systems programming concepts to enhance scalability, improving system reliability and reducing message delivery failures by 30%

Technologies and Tools: WebRTC, C++, DevOps, C, Java, JNI, Elastic Search, Logstash, Agile, Waterfall, Jira, Git, CI/CD

PROJECT EXPERIENCE

Software Developer | University of Texas at Dallas | Dallas TX

Sep 2024 – May 2025

Grader Assignment System: Grader Assignment System is a web application used by UT Dallas to hire graders for CS courses each semester

- Translated project goals into detailed REST API specifications and database schemas, designing Grader Assignment System Web UI effectively
- Mentored 12 student teams in Agile, Scrum, and Waterfall, leveraging Jira while implementing AI-powered LLM and GenAI solutions

Multiplayer Snake Game: The Multiplayer Snake Game is a Java-based application, inspired by Nokia Snake, enhanced with real-time multiplayer functionality

- Designed and implemented a React-based UI and multithreaded game engine, achieving low-latency real-time multiplayer gameplay
- Expanded automated test scripts with unit, regression, and integration testing, ensuring network protocol accuracy and server reliability

Context Dependent Natural Language to SQL Query Generator: CDNL to SQL Query Generator is an AI system that converts natural language queries into SQL, enabling intuitive database interactions

- Improved and deployed scalable backend REST APIs using Flask and Python for LLM-powered natural language database query processing
- Built robust graph-based schema pipelines and deployed services on Google Cloud, enabling complex queries and readiness

Technologies and Tools: REST API, Java, Spring Boot, Python, Git, API, SQL, Google Cloud Platform, Jira, Gemini, ChatGPT, LLMs, Machine Learning, Artificial Intelligence, Natural Language Processing, PyTorch, Tensorflow

CERTIFICATIONS

- Coursera Certification, Networking Specialization by CISCO
- NPTEL Certification, Programming in Java, IIT Kharagpur
- Coursera Certification, Python Specialization, University of Michigan

PERSONAL ACCOMPLISHMENTS

- Recognized by senior leadership for identifying and fixing a critical AV framework bug, significantly improving Zoho's platform stability
- Awarded "Game Changer of the Year – 2021" at Zoho for delivering high-impact technical improvements and product enhancements

TECHNICAL SKILLS

- Frameworks & UI:** React, Express.js, Node.js, Spring Boot, Figma | **Containerization & Orchestration:** Docker, Kubernetes
- Databases:** MySQL, PostgreSQL, Oracle, MongoDB, Cassandra | **Machine Learning Tools:** TensorFlow, PyTorch, Kubeflow
- Cloud Platforms:** AWS, Azure, Google Cloud Platform, Oracle Cloud | **Operating Systems:** Linux, CentOS, Debian, RHEL
- DevOps & Tools:** Git, GitHub, Jenkins, Jira, CI/CD, Bash, Visual Studio, Vim | **Languages:** C++, C, Java, Python

CORE COMPETENCIES

Real-time distributed systems & backend development | Cloud-native application design & deployment | Microservices architecture & system scalability | DevOps, CI/CD pipelines, and infrastructure optimization | Low-latency algorithms for high-traffic applications | AI/ML model integration and intelligent query systems | Cross-functional collaboration & technical leadership | System architecture & performance optimization | API development & database schema design | Problem-solving & root-cause analysis | Agile, Scrum, and Waterfall project execution | Debugging, troubleshooting & production issue resolution | Scalable design patterns & resilience engineering | Leadership & mentoring | Effective communication & stakeholder collaboration